



acoustics

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```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1      ✓ readr      2.2.0
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2     4.0.3      ✓ tibble     3.3.1
✓ lubridate  1.9.5      ✓ tidyr      1.3.2
✓ purrr       1.2.2
— Conflicts — tidyverse_conflicts()
—
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

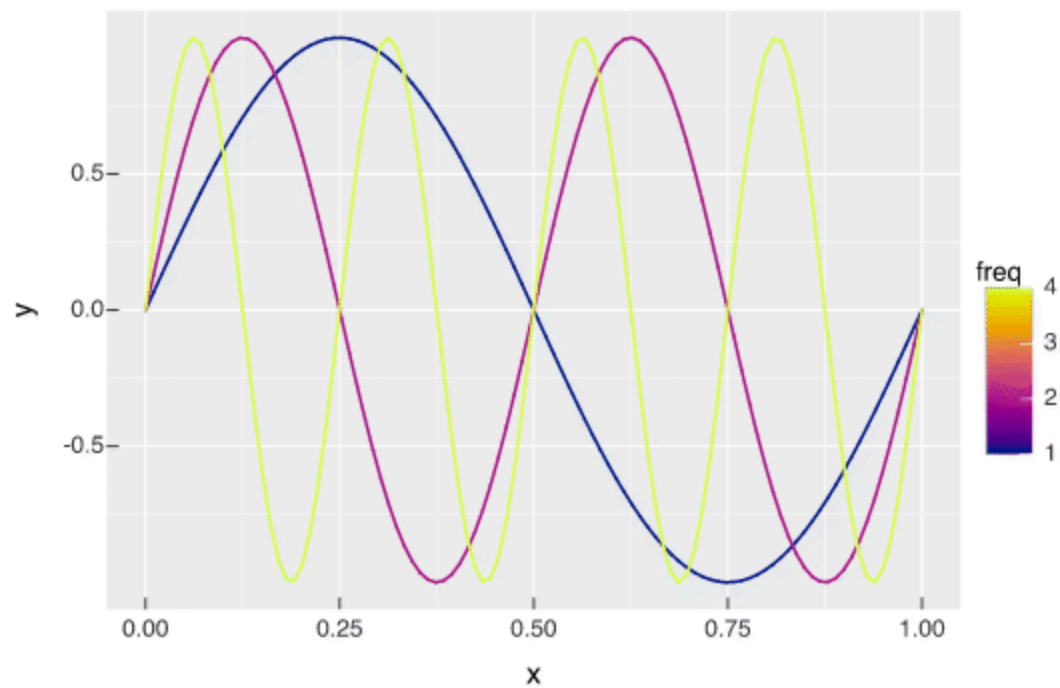
```
box::use(
  v = vellumplot
)
```

```
make_sine <- function(x, freq = 1){
  # if there's no `freq` column
  # set `freq` to 1.
  y = sin(
    (freq * x * 2 * pi)
  )
}
```

```
make_moving_sine <- function(x, freq, t){
  sin(
    (freq * (x - t) * 2 * pi)
  )
}
```



```
expand_grid(
  x = seq(0, 1, length = 100),
  freq = c(1, 2, 4),
  t = seq(0, 3, length = 100)
) |>
mutate(
  .by = freq,
  y = make_moving_sine(x, freq, t)
) |>
v$vpplot() |>
v$mark_line(
  x = x,
  y = y,
  color = freq
) |>
v$transition_time(
  t
) |>
v$scale_color_continuous(
  palette = "Plasma"
) |>
v$animate(
  nframes = 200
) |>
v$anim_save(
  filename = "test.gif"
)
```



```
make_moving_disp <- function(x, freq=1, t=0){  
  cos(  
    (freq * (x - t)* 2 * pi)  
  )  
}
```

```
make_disp_df <- function(df, freq=1, t=0){  
  df |>  
    mutate(  
      x = x + (make_moving_disp(  
        x, freq, t  
      ) * 0.09),  
      disp = make_moving_disp(  
        x, freq, t  
      ),  
      t = t  
    )  
}
```

```
tibble(  
  x = runif(10000, max = 2),
```



```
y = runif(10000)
) |>
mutate(
  id = row_number()
)->
air

air |>
  filter(
    between(x, 0.99, 1.1)
  ) |>
pull(id) |>
sample(20) ->
chosen

air |>
  mutate(
    mark = id %in% chosen
  ) ->
air

map(
  seq(0, 2.5*pi, length = 100),
  ~make_disp_df(
    df = air,
    freq = 1,
    t = .x
  )
) |>
list_rbind() ->
big_df
```

```
big_df |>
v$vpplot() |>
v$mark_point(
  x = x,
  y = y,
  size = 0.5,
  color = mark
) |>
v$scale_color_manual(
  values = c("black", "red")
) |>
```



```
v$guides(  
  color = v$guide_none()  
) |>  
v$scale_x_continuous(  
  expand = 0  
) |>  
v$scale_y_continuous(  
  expand = 0  
) |>  
v$transition_time(  
  t  
) |>  
v$ease_aes("cubic-in-out") |>  
v$theme_void() |>  
v$animate(nframes = 200, fps = 12) |>  
v$anim_save(  
  filename = "air.gif"  
)
```

